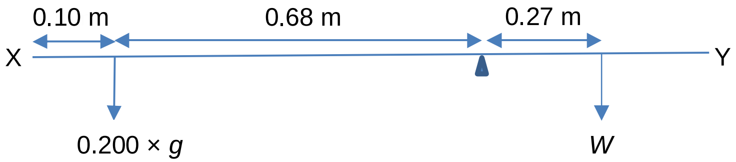
5. Answer: A

When broom is moved rightwards by 0.27m, the chair stays at the same position.
Hence the pivot is shifted further from Y by 0.27m.

Perpendicular distance from 200g mass to new position of pivot = 1.05-0.10-0.27 =0.68m



Take moment about the pivot,
Total clockwise moment = Total anticlockwise moment

$$W \times 0.27 = 0.200 \times 9.81 \times 0.68$$
$$W = 4.941 \text{ N}$$

Hence the mass of the broom = $\dfrac{4.941}{9.81}$
= 0.504 kg
= 504 g

6. Answer: A